

Appendices

Appendix A - Table 1. Samples and metadata

Sample_name	BioProject	Product	Ripenin g	Geo	Host_genome	PGI_area
C03_1	PRJNA997800	cecina	7m	Spain: León	<i>Bos taurus</i> ARS-UCD1.2	
C03_2	PRJNA997800	cecina	7m	Spain: León	<i>Bos taurus</i> ARS-UCD1.2	
C06_1	PRJNA997800	cecina	9m	Spain: León	<i>Bos taurus</i> ARS-UCD1.2	
C06_2	PRJNA997800	cecina	9m	Spain: León	<i>Bos taurus</i> ARS-UCD1.2	
C01_1	PRJNA997800	cecina	~12m	Spain: León	<i>Bos taurus</i> ARS-UCD1.2	
C01_2	PRJNA997800	cecina	~12m	Spain: León	<i>Bos taurus</i> ARS-UCD1.2	
C02_1	PRJNA997800	cecina	~9m	Spain: León	<i>Bos taurus</i> ARS-UCD1.2	
C02_2	PRJNA997800	cecina	~9m	Spain: León	<i>Bos taurus</i> ARS-UCD1.2	
C04_1	PRJNA997800	cecina	~9m	Spain: León	<i>Bos taurus</i> ARS-UCD1.2	
C04_2	PRJNA997800	cecina	~9m	Spain: León	<i>Bos taurus</i> ARS-UCD1.2	
C05_1	PRJNA997800	cured tongue	1m	Spain: León	<i>Bos taurus</i> ARS-UCD1.2	
C05_2	PRJNA997800	cured tongue	1m	Spain: León	<i>Bos taurus</i> ARS-UCD1.2	
C07_1	PRJNA997800	dry-aged beef	2m	Spain: León	<i>Bos taurus</i> ARS-UCD1.2	
C07_2	PRJNA997800	dry-aged beef	2m	Spain: León	<i>Bos taurus</i> ARS-UCD1.2	
C08_1	PRJNA997800	dry-aged beef	>3m	Spain: León	<i>Bos taurus</i> ARS-UCD1.2	
C08_2	PRJNA997800	dry-aged beef	>3m	Spain: León	<i>Bos taurus</i> ARS-UCD1.2	
F01_1	PRJNA997800	DFS chorizo	45d	Spain: León	<i>Sus scrofa</i> Sscrofa11.1	
F01_2	PRJNA997800	DFS chorizo	45d	Spain: León	<i>Sus scrofa</i> Sscrofa11.1	
F02_1	PRJNA997800	DFS chorizo	35d	Spain: León	<i>Sus scrofa</i> Sscrofa11.1	
F02_2	PRJNA997800	DFS chorizo	35d	Spain: León	<i>Sus scrofa</i> Sscrofa11.1	
F03_1	PRJNA997800	DFS chorizo	>1m	Spain: León	<i>Sus scrofa</i> Sscrofa11.1	
F03_2	PRJNA997800	DFS chorizo	>1m	Spain: León	<i>Sus scrofa</i> Sscrofa11.1	
F04_1	PRJNA997800	DFS chorizo	25d	Spain: León	<i>Sus scrofa</i> Sscrofa11.1	
F04_2	PRJNA997800	DFS chorizo	25d	Spain: León	<i>Sus scrofa</i> Sscrofa11.1	
F05_1	PRJNA997800	DFS salchichon	45d	Spain: León	<i>Sus scrofa</i> Sscrofa11.1	
F05_2	PRJNA997800	DFS salchichon	45d	Spain: León	<i>Sus scrofa</i> Sscrofa11.1	
F06_1	PRJNA997800	DFS chorizo	1m	Spain: León	<i>Sus scrofa</i> Sscrofa11.1	
F06_2	PRJNA997800	DFS chorizo	1m	Spain: León	<i>Sus scrofa</i> Sscrofa11.1	
F07_1	PRJNA997800	DFS salchichon	>1m	Spain: León	<i>Sus scrofa</i> Sscrofa11.1	
F07_2	PRJNA997800	DFS salchichon	>1m	Spain: León	<i>Sus scrofa</i> Sscrofa11.1	
C0_1_MSO	PRJNA1337460	Miao-style sour meat (suan rou)	0d	China:Hu Nan	<i>Sus scrofa</i> Sscrofa11.1 + <i>Oryza sativa</i> IRGSP-1.0 (rice)	
C0_2_MSO	PRJNA1337460	Miao-style sour meat (suan rou)	0d	China:Hu Nan	<i>Sus scrofa</i> Sscrofa11.1 + <i>Oryza sativa</i> IRGSP-1.0 (rice)	
C0_3_MSO	PRJNA1337460	Miao-style sour meat (suan rou)	0d	China:Hu Nan	<i>Sus scrofa</i> Sscrofa11.1 + <i>Oryza sativa</i> IRGSP-1.0 (rice)	
C15_1_MSO	PRJNA1337460	Miao-style sour meat (suan rou)	15d	China:Hu Nan	<i>Sus scrofa</i> Sscrofa11.1 + <i>Oryza sativa</i> IRGSP-1.0 (rice)	

Sample_name	BioProject	Product	Ripenin g	Geo	Host_genome	PGI_area
C15_2_MSO	PRJNA1337460	Miao-style sour meat (suan rou)	15d	China:Hu Nan	<i>Sus scrofa</i> Sscrofa11.1 + <i>Oryza sativa</i> IRGSP-1.0 (rice)	
C15_3_MSO	PRJNA1337460	Miao-style sour meat (suan rou)	15d	China:Hu Nan	<i>Sus scrofa</i> Sscrofa11.1 + <i>Oryza sativa</i> IRGSP-1.0 (rice)	
C30_1_MSO	PRJNA1337460	Miao-style sour meat (suan rou)	30d	China:Hu Nan	<i>Sus scrofa</i> Sscrofa11.1 + <i>Oryza sativa</i> IRGSP-1.0 (rice)	
C30_2_MSO	PRJNA1337460	Miao-style sour meat (suan rou)	30d	China:Hu Nan	<i>Sus scrofa</i> Sscrofa11.1 + <i>Oryza sativa</i> IRGSP-1.0 (rice)	
C30_3_MSO	PRJNA1337460	Miao-style sour meat (suan rou)	30d	China:Hu Nan	<i>Sus scrofa</i> Sscrofa11.1 + <i>Oryza sativa</i> IRGSP-1.0 (rice)	
C45_1_MSO	PRJNA1337460	Miao-style sour meat (suan rou)	45d	China:Hu Nan	<i>Sus scrofa</i> Sscrofa11.1 + <i>Oryza sativa</i> IRGSP-1.0 (rice)	
C45_2_MSO	PRJNA1337460	Miao-style sour meat (suan rou)	45d	China:Hu Nan	<i>Sus scrofa</i> Sscrofa11.1 + <i>Oryza sativa</i> IRGSP-1.0 (rice)	
C45_3_MSO	PRJNA1337460	Miao-style sour meat (suan rou)	45d	China:Hu Nan	<i>Sus scrofa</i> Sscrofa11.1 + <i>Oryza sativa</i> IRGSP-1.0 (rice)	
C60_1_MSO	PRJNA1337460	Miao-style sour meat (suan rou)	60d	China:Hu Nan	<i>Sus scrofa</i> Sscrofa11.1 + <i>Oryza sativa</i> IRGSP-1.0 (rice)	
C60_2_MSO	PRJNA1337460	Miao-style sour meat (suan rou)	60d	China:Hu Nan	<i>Sus scrofa</i> Sscrofa11.1 + <i>Oryza sativa</i> IRGSP-1.0 (rice)	
C60_3_MSO	PRJNA1337460	Miao-style sour meat (suan rou)	60d	China:Hu Nan	<i>Sus scrofa</i> Sscrofa11.1 + <i>Oryza sativa</i> IRGSP-1.0 (rice)	
Y0_1_MSO	PRJNA1337460	Miao-style sour meat (suan rou)	0d	China:Hu Nan	<i>Sus scrofa</i> Sscrofa11.1 + <i>Oryza sativa</i> IRGSP-1.0 (rice)	
Y0_2_MSO	PRJNA1337460	Miao-style sour meat (suan rou)	0d	China:Hu Nan	<i>Sus scrofa</i> Sscrofa11.1 + <i>Oryza sativa</i> IRGSP-1.0 (rice)	
Y0_3_MSO	PRJNA1337460	Miao-style sour meat (suan rou)	0d	China:Hu Nan	<i>Sus scrofa</i> Sscrofa11.1 + <i>Oryza sativa</i> IRGSP-1.0 (rice)	
Y15_1_MSO	PRJNA1337460	Miao-style sour meat (suan rou)	15d	China:Hu Nan	<i>Sus scrofa</i> Sscrofa11.1 + <i>Oryza sativa</i> IRGSP-1.0 (rice)	
Y15_2_MSO	PRJNA1337460	Miao-style sour meat (suan rou)	15d	China:Hu Nan	<i>Sus scrofa</i> Sscrofa11.1 + <i>Oryza sativa</i> IRGSP-1.0 (rice)	
Y15_3_MSO	PRJNA1337460	Miao-style sour meat (suan rou)	15d	China:Hu Nan	<i>Sus scrofa</i> Sscrofa11.1 + <i>Oryza sativa</i> IRGSP-1.0 (rice)	
Y30_1_MSO	PRJNA1337460	Miao-style sour meat (suan rou)	30d	China:Hu Nan	<i>Sus scrofa</i> Sscrofa11.1 + <i>Oryza sativa</i> IRGSP-1.0 (rice)	
Y30_2_MSO	PRJNA1337460	Miao-style sour meat (suan rou)	30d	China:Hu Nan	<i>Sus scrofa</i> Sscrofa11.1 + <i>Oryza sativa</i> IRGSP-1.0 (rice)	
Y30_3_MSO	PRJNA1337460	Miao-style sour meat (suan rou)	30d	China:Hu Nan	<i>Sus scrofa</i> Sscrofa11.1 + <i>Oryza sativa</i> IRGSP-1.0 (rice)	
Y45_1_MSO	PRJNA1337460	Miao-style sour meat (suan rou)	45d	China:Hu Nan	<i>Sus scrofa</i> Sscrofa11.1 + <i>Oryza sativa</i> IRGSP-1.0 (rice)	

Sample_name	BioProject	Product	Ripenin g	Geo	Host_genome	PGI_area
Y45_2_MSO	PRJNA1337460	Miao-style sour meat (suan rou)	45d	China:Hu Nan	<i>Sus scrofa</i> Sscrofa11.1 + <i>Oryza sativa</i> IRGSP-1.0 (rice)	
Y45_3_MSO	PRJNA1337460	Miao-style sour meat (suan rou)	45d	China:Hu Nan	<i>Sus scrofa</i> Sscrofa11.1 + <i>Oryza sativa</i> IRGSP-1.0 (rice)	
Y60_1_MSO	PRJNA1337460	Miao-style sour meat (suan rou)	60d	China:Hu Nan	<i>Sus scrofa</i> Sscrofa11.1 + <i>Oryza sativa</i> IRGSP-1.0 (rice)	
Y60_2_MSO	PRJNA1337460	Miao-style sour meat (suan rou)	60d	China:Hu Nan	<i>Sus scrofa</i> Sscrofa11.1 + <i>Oryza sativa</i> IRGSP-1.0 (rice)	
Y60_3_MSO	PRJNA1337460	Miao-style sour meat (suan rou)	60d	China:Hu Nan	<i>Sus scrofa</i> Sscrofa11.1 + <i>Oryza sativa</i> IRGSP-1.0 (rice)	
CS_1_MS	PRJNA895928	Fermented mutton sausage	NA	China: Chengdu	<i>Ovis aries</i> ARS-UI_Ramb_v2.0	
CS_2_MS	PRJNA895928	Fermented mutton sausage	NA	China: Chengdu	<i>Ovis aries</i> ARS-UI_Ramb_v2.0	
CS_3_MS	PRJNA895928	Fermented mutton sausage	NA	China: Chengdu	<i>Ovis aries</i> ARS-UI_Ramb_v2.0	
FJ_1_MS	PRJNA895928	Fermented mutton sausage	NA	China: Chengdu	<i>Ovis aries</i> ARS-UI_Ramb_v2.0	
FJ_2_MS	PRJNA895928	Fermented mutton sausage	NA	China: Chengdu	<i>Ovis aries</i> ARS-UI_Ramb_v2.0	
FJ_3_MS	PRJNA895928	Fermented mutton sausage	NA	China: Chengdu	<i>Ovis aries</i> ARS-UI_Ramb_v2.0	
GZ_2_MS	PRJNA895928	Fermented mutton sausage	NA	China: Chengdu	<i>Ovis aries</i> ARS-UI_Ramb_v2.0	
GZ_3_MS	PRJNA895928	Fermented mutton sausage	NA	China: Chengdu	<i>Ovis aries</i> ARS-UI_Ramb_v2.0	
YZ_1_MS	PRJNA895928	Fermented mutton sausage	NA	China: Chengdu	<i>Ovis aries</i> ARS-UI_Ramb_v2.0	
YZ_2_MS	PRJNA895928	Fermented mutton sausage	NA	China: Chengdu	<i>Ovis aries</i> ARS-UI_Ramb_v2.0	
YZ_3_MS	PRJNA895928	Fermented mutton sausage	NA	China: Chengdu	<i>Ovis aries</i> ARS-UI_Ramb_v2.0	
SP2BYd15	PRJNA636619	Salame Piemonte PGI (Italian dry-fermented salami)	15d	Piedmont (Piemonte), Turin area, Italy	<i>Sus scrofa</i> Sscrofa11.1	
SP2BXd15	PRJNA636619	Salame Piemonte PGI (Italian dry-fermented salami)	15d	Piedmont (Piemonte), Turin area, Italy	<i>Sus scrofa</i> Sscrofa11.1	
SP2BZd15	PRJNA636619	Salame Piemonte PGI (Italian dry-fermented salami)	15d	Piedmont (Piemonte), Turin area, Italy	<i>Sus scrofa</i> Sscrofa11.1	
SP2BYd30	PRJNA636619	Salame Piemonte PGI (Italian dry-fermented salami)	30d	Piedmont (Piemonte), Turin area, Italy	<i>Sus scrofa</i> Sscrofa11.1	
SP2BXd30	PRJNA636619	Salame Piemonte PGI (Italian dry-fermented salami)	30d	Piedmont (Piemonte), Turin area, Italy	<i>Sus scrofa</i> Sscrofa11.1	
SP2BZd30	PRJNA636619	Salame Piemonte PGI (Italian dry-fermented salami)	30d	Piedmont (Piemonte), Turin area, Italy	<i>Sus scrofa</i> Sscrofa11.1	
SP2BYd4	PRJNA636619	Salame Piemonte PGI (Italian dry-fermented salami)	4d	Piedmont (Piemonte), Turin area, Italy	<i>Sus scrofa</i> Sscrofa11.1	
SP2BXd4	PRJNA636619	Salame Piemonte PGI (Italian dry-fermented salami)	4d	Piedmont (Piemonte), Turin area, Italy	<i>Sus scrofa</i> Sscrofa11.1	
SP2BZd4	PRJNA636619	Salame Piemonte PGI (Italian dry-fermented salami)	4d	Piedmont (Piemonte), Turin area, Italy	<i>Sus scrofa</i> Sscrofa11.1	
SP2BYd50	PRJNA636619	Salame Piemonte PGI (Italian dry-fermented salami)	50d	Piedmont (Piemonte), Turin area, Italy	<i>Sus scrofa</i> Sscrofa11.1	
SP2BXd50	PRJNA636619	Salame Piemonte PGI (Italian dry-fermented salami)	50d	Piedmont (Piemonte), Turin area, Italy	<i>Sus scrofa</i> Sscrofa11.1	

Sample_name	BioProject	Product	Ripenin g	Geo	Host_genome	PGI_area
SP2BZd50	PRJNA636619	Salame Piemonte PGI (Italian dry-fermented salami)	50d	Piedmont (Piemonte), Turin area, Italy	<i>Sus scrofa</i> Sscrofa11.1	
SP2BYd8	PRJNA636619	Salame Piemonte PGI (Italian dry-fermented salami)	8d	Piedmont (Piemonte), Turin area, Italy	<i>Sus scrofa</i> Sscrofa11.1	
SP2BXd8	PRJNA636619	Salame Piemonte PGI (Italian dry-fermented salami)	8d	Piedmont (Piemonte), Turin area, Italy	<i>Sus scrofa</i> Sscrofa11.1	
SP2BZd8	PRJNA636619	Salame Piemonte PGI (Italian dry-fermented salami)	8d	Piedmont (Piemonte), Turin area, Italy	<i>Sus scrofa</i> Sscrofa11.1	
SF3RAt0	PRJNA352236	Salame Felino-type Italian dry-fermented salami	0d	Italy	<i>Sus scrofa</i> Sscrofa11.1	
SF3RBt0	PRJNA352236	Salame Felino-type Italian dry-fermented salami	0d	Italy	<i>Sus scrofa</i> Sscrofa11.1	
SF3RAt3	PRJNA352236	Salame Felino-type Italian dry-fermented salami	3d	Italy	<i>Sus scrofa</i> Sscrofa11.1	
SF3RBt3	PRJNA352236	Salame Felino-type Italian dry-fermented salami	3d	Italy	<i>Sus scrofa</i> Sscrofa11.1	
SF3RAt40	PRJNA352236	Salame Felino-type Italian dry-fermented salami	40d	Italy	<i>Sus scrofa</i> Sscrofa11.1	
SF3RBt40	PRJNA352236	Salame Felino-type Italian dry-fermented salami	40d	Italy	<i>Sus scrofa</i> Sscrofa11.1	
SF3RAt7	PRJNA352236	Salame Felino-type Italian dry-fermented salami	7d	Italy	<i>Sus scrofa</i> Sscrofa11.1	
SF3RBt7	PRJNA352236	Salame Felino-type Italian dry-fermented salami	7d	Italy	<i>Sus scrofa</i> Sscrofa11.1	
SAP1		Ciauscolo PGI	72d	Italy	<i>Sus scrofa</i> Sscrofa11.1	within PGI area (Ascoli Piceno); reported biological replicate A
SAP2		Ciauscolo PGI	72d	Italy	<i>Sus scrofa</i> Sscrofa11.1	within PGI area (Ascoli Piceno); reported biological replicate B
SMC1		Ciauscolo PGI	44d	Italy	<i>Sus scrofa</i> Sscrofa11.1	within PGI area (Macerata); reported biological replicate A
SMC2		Ciauscolo PGI	44d	Italy	<i>Sus scrofa</i> Sscrofa11.1	within PGI area (Macerata); reported biological replicate B
SPG1		Ciauscolo-type product (outside PGI area)	NA	Italy	<i>Sus scrofa</i> Sscrofa11.1	outside PGI area (Perugia, Umbria); reported biological replicate A
SPG2		Ciauscolo-type product (outside PGI area)	NA	Italy	<i>Sus scrofa</i> Sscrofa11.1	outside PGI area (Perugia, Umbria); reported biological replicate B

Appendix B - Table 2. Software versions and parameters

The raw reproducibility workbook is available on request.

Category	Resource	Version	Source / environment	Identifier and notes
Deposited data	Metagenomes - beef and pork DFS		Barcenilla et al. 2024	See raw reproducibility workbook for detailed provenance and run notes.
Deposited data	Metagenomes - Salame Felino		Ferrocino et al. 2018	See raw reproducibility workbook for detailed provenance and run notes.
Deposited data	Metagenomes - Salame Piemonte		Franciosa et al. 2021	See raw reproducibility workbook for detailed provenance and run notes.
Deposited data	Metagenomes - Mutton sausage		BioProject PRJNA895928 (no associated publication identified)	See raw reproducibility workbook for detailed provenance and run notes.
Deposited data	Metagenomes - Miao suan rou (inoculated external group)		BioProject PRJNA1337460	See raw reproducibility workbook for detailed provenance and run notes.
Deposited data	Ciauscolo MUST_25 (dataset)		This study	See raw reproducibility workbook for detailed provenance and run notes.
Software - preparation	sra-tools (prefetch, fasterq-dump)	3.2.1	env ncbi	See raw reproducibility workbook for detailed provenance and run notes.
Software - preparation	fastp	1.3.0	env trim	See raw reproducibility workbook for detailed provenance and run notes.
Software - preparation	FastQC	0.12.1	env qc	See raw reproducibility workbook for detailed provenance and run notes.
Software - preparation	MultiQC	1.33	env qc	See raw reproducibility workbook for detailed provenance and run notes.
Software - preparation	Bowtie2	2.5.5	env host_filter	See raw reproducibility workbook for detailed provenance and run notes.
Software - preparation	samtools	1.23.1	env binning / host_filter	See raw reproducibility workbook for detailed provenance and run notes.
Software - profiling	MetaPhlAn	4.2.4 (21 Oct 2025)	env profiling	mpa_vJan25_CHOCOPhlanSGB_202503 database; one sample unclassified
Software - profiling	StrainPhlAn	4.2.4 (21 Oct 2025)	env profiling	See raw reproducibility workbook for detailed provenance and run notes.
Software - profiling	Kraken2	2.17.1	env kraken2	Confidence threshold: 0.4
Software - profiling	Bracken	3.1	env kraken2	Read length 150 bp; minimum 10 reads per species
Software - profiling	sylph	0.9.0	env benchmark	Abundance-corrected MinHash sketch cross-check
Software - assembly and binning	metaSPAdes	4.2.0	env assembly	See raw reproducibility workbook for detailed provenance and run notes.
Software - assembly and binning	MEGAHIT	1.2.9	env benchmark	See raw reproducibility workbook for detailed provenance and run notes.
Software - assembly and binning	MetaBAT2	2.18	env binning	See raw reproducibility workbook for detailed provenance and run notes.
Software - assembly and binning	EukRep	0.6.7	env eukrep	scikit-learn 0.19 pinned
Software - genome quality control	CheckM2	1.1.0	env checkm2	See raw reproducibility workbook for detailed provenance and run notes.
Software - genome quality control	EukCC	2.1.2	env eukcc	See raw reproducibility workbook for detailed provenance and run notes.
Software - genome quality control	GUNC	1.1.1	env gunc	proGenomes 2.1 database; CSS threshold 0.45

Category	Resource	Version	Source / environment	Identifier and notes
Software - genome quality control	dRep	3.7.1	env drep	Secondary ANI 0.95 (species); primary ANI 0.90
Software - taxonomy	GTDB-Tk	2.7.2	env gtdbtk	See raw reproducibility workbook for detailed provenance and run notes.
Software - taxonomy	FastANI	1.34	env drep (run) / env fastani / env gtdbtk	Aligned-fraction threshold >=0.50; version verified from run log
Software - presence and strains	CoverM	0.8.0	env coverm	Genome breadth >=50%; read-mapping detection
Software - presence and strains	inStrain	1.3.9	env instrain	Competitive mapping; percent_compared ≥0.50; popANI ≥0.99999; --database_mode not used
Software - presence and strains	PanPhlAn	3.1	env panphlan	STRICT: min_coverage 2; left_max 1.25; right_min 0.75
Software - phylogeny	RAxML	8.2.13	env profiling	Marker tree: 182 references and 47 sample consensus sequences
Software - phylogeny	IQ-TREE	3.1.1	env profiling	VT+F+ASC+R6; 1,000 SH-aLRT and 1,000 ultrafast bootstrap replicates
Software - phylogeny	PhyloPhlAn	3.2.1	env profiling	Whole-genome alignment: 18,199 amino-acid positions
Software - phylogeny	DIAMOND (within PhyloPhlAn)	2.1.24	env profiling	PhyloPhlAn internal search settings recorded in raw workbook
Software - phylogeny	MAFFT (within PhyloPhlAn)	7.526 (2024/Apr/26)	env profiling	PhyloPhlAn internal alignment settings recorded in raw workbook
Software - phylogeny	trimAl (within PhyloPhlAn)	1.5.1 (reported by the program as v1.5.rev1, build 2025-11-25)	env profiling	gappyout trimming
Software - phylogeny	FastTreeMP (within PhyloPhlAn)	2.2.0	env profiling	SH-like local supports; not bootstrap
Software - phylogeny	PhiPack	1.1	env phipack	100 evaluable markers per panel (99 shared); 1,000 permutations; nominal p≤0.05; Bonferroni threshold p≤0.0005 not assessable at this permutation resolution
Statistical environment	R	4.5.2	archived sessionInfo	vegan 2.7-3; permute 0.9-10; rstatix 0.7.3; ade4 1.7-24. PERMDISP version not separately logged.

Appendix C.1 - Table 3A. Prokaryotic representatives

MAG_representative	Assigned_taxon_GTDB	ANI_%	Completeness_%	Contamination_%	In_core_wildtype	Product(s)
cured_tongue.9	<i>Staphylococcus saprophyticus</i>	99.68	78.4	0.14	YES	cured_tongue/miao_suanrou/mutton_sausage
cured_tongue.20	<i>Staphylococcus equorum_A</i>	98.58	99.22	0.18	YES	DFS_salchichon/cured_tongue
felino_salami.8	<i>Staphylococcus equorum</i>	98.97	92.22	0.17	YES	cecina/dry_aged_beef/felino_salami/miao_suanrou/mutton_sausage
dry_aged_beef.33	<i>Staphylococcus succinus</i>	95.18	88.96	0.15	YES	dry_aged_beef
felino_salami.15	<i>Staphylococcus xylosus</i>	97.77	96.75	1.29	YES	felino_salami
miao_suanrou.101	<i>Staphylococcus kloosii</i>	99	86.18	1.67	NO	miao_suanrou
miao_suanrou.8	<i>Macroccoides psychrotolerans</i>	99.28	99.88	7.54	YES (via wild-type cluster member)	felino_salami/miao_suanrou

MAG_representative	Assigned_taxon_GTDB	ANI_%	Completeness_%	Contamination_%	In_core_wildtype	Product(s)
miao_suanrou.115	<i>Macrococcoides bohemicum</i>	98.44	90.5	0.55	NO	miao_suanrou
miao_suanrou.99	<i>Mammaliicoccus vitulinus</i>	99.34	99.92	2.42	NO	miao_suanrou
miao_suanrou.123	<i>Weissella</i> sp. (uSGB)	N/A	74.96	0.25	NO	miao_suanrou
miao_suanrou.68	<i>Weissella</i> sp. (uSGB)	N/A	74.74	3.49	NO	miao_suanrou
miao_suanrou.84	<i>Weissella ceti</i>	98.9	80.76	8.31	NO	miao_suanrou
miao_suanrou.75	<i>Weissella hellenica</i>	99.03	78.66	2.82	NO	miao_suanrou
miao_suanrou.94	<i>Weissella minor</i>	99.06	64.03	0.57	NO	miao_suanrou
miao_suanrou.129	<i>Weissella cibaria</i>	98.75	84.98	3.11	NO	miao_suanrou
miao_suanrou.72	<i>Weissella paramesenteroides</i>	98.56	86.66	1.59	NO	miao_suanrou
dry_aged_beef.4	<i>Enterococcus_D casseliflavus</i>	98.85	94.67	1.22	YES	DFS_chorizo/dry_aged_beef
miao_suanrou.116	<i>Enterococcus_D</i> sp001297065	97.67	99.97	1.33	NO	miao_suanrou
miao_suanrou.125	<i>Enterococcus_A viikkiensis</i>	97.91	57.23	1.03	NO	miao_suanrou
miao_suanrou.21	<i>Enterococcus_A hermanniensis</i>	99.83	80.16	1.04	NO	miao_suanrou
miao_suanrou.82	<i>Enterococcus faecalis</i>	98.66	98.73	6.25	NO	miao_suanrou
miao_suanrou.61	<i>Carnobacterium maltaromaticum</i>	99.3	94.7	0.57	NO	miao_suanrou
miao_suanrou.83	<i>Carnobacterium divergens</i>	99.11	97.87	4.54	NO	miao_suanrou
miao_suanrou.23	<i>Enterococcus_G</i> sp. (uSGB)	N/A	86.24	0.84	NO	miao_suanrou
miao_suanrou.24	<i>Lentilactobacillus parabuchneri</i>	99.67	92.45	0.32	NO	miao_suanrou
miao_suanrou.80	<i>Lentilactobacillus buchneri</i>	99.75	89.09	5.57	NO	miao_suanrou
miao_suanrou.74	<i>Lentilactobacillus kisonensis</i>	99.8	84.92	0.7	NO	miao_suanrou
miao_suanrou.58	<i>Lentilactobacillus parafarraginis</i>	98.87	92.46	3.89	NO	miao_suanrou
piemonte_salami.23	<i>Pediococcus acidilactici</i>	97.14	95.43	0.01	YES	miao_suanrou/piemonte_salami
miao_suanrou.89	<i>Pediococcus pentosaceus</i>	99.11	95.67	5.23	NO	miao_suanrou
DFS_salchichon.21	<i>Levilactobacillus brevis</i>	96.93	99.99	0.19	YES	DFS_salchichon/felino_salami/miao_suanrou/piemonte_salami
DFS_chorizo.50	<i>Lactiplantibacillus paraplantarum</i>	98.05	78.04	9.83	YES	DFS_chorizo
DFS_chorizo.45	<i>Levilactobacillus cerevisiae</i>	95.7	98.71	3.3	YES	DFS_chorizo
dry_aged_beef.30	<i>Brochothrix thermosphacta</i>	99.5	98.61	0.05	YES	DFS_chorizo/cecina/dry_aged_beef/felino_salami/miao_suanrou
piemonte_salami.24	<i>Loigolactobacillus coryniformis</i>	98.2	86.39	1.26	YES	piemonte_salami
cured_tongue.28	<i>Latilactobacillus sakei</i>	97.46	99.65	0.36	YES	DFS_salchichon/ciauscolo/cured_tongue/felino_salami/miao_suanrou/piemonte_salami
ciauscolo.5	<i>Latilactobacillus curvatus</i>	97.69	87.06	0.21	YES	DFS_chorizo/DFS_salchichon/ciauscolo/felino_salami/piemonte_salami
ciauscolo.4	<i>Lactococcus lactis</i>	99.12	100	0.58	YES	ciauscolo
miao_suanrou.25	<i>Lactococcus cremoris</i>	98.04	82.43	1.34	NO	miao_suanrou
DFS_salchichon.14	<i>Streptococcus thermophilus</i>	98.72	93.55	0.17	YES	DFS_salchichon
miao_suanrou.32	<i>Pseudolactococcus raffinolactis</i>	98.86	96.28	0.15	NO	miao_suanrou
DFS_salchichon.18	<i>Dellagliaa kimchii</i>	97.99	99.98	0.08	YES	DFS_chorizo/DFS_salchichon
cured_tongue.5	<i>Tetragenococcus</i> sp030462125	98.38	99.13	0.19	YES	cured_tongue
felino_salami.20	<i>Lactococcus formosensis</i>	97.77	83.03	3.41	YES	felino_salami
felino_salami.7	<i>Leuconostoc mesenteroides</i>	99.42	95.37	2.06	YES	felino_salami/miao_suanrou

MAG_representative	Assigned_taxon_GTDB	ANI_%	Completeness_%	Contamination_%	In_core_wildtype	Product(s)
miao_suanrou.113	<i>Pediococcus sp. (uSGB)</i>	N/A	71.39	0.04	NO	miao_suanrou
piemonte_salami.18	<i>Leuconostoc carnosum</i>	99.94	99.93	0.06	YES	felino_salami/piemonte_salami
DFS_chorizo.48	<i>Leuconostoc gelidum</i>	99.54	93.47	5.83	YES	DFS_chorizo/miao_suanrou
felino_salami.12	<i>Leuconostoc gasicomitatum</i>	98.82	94.78	7.47	YES	felino_salami
felino_salami.5	<i>Leuconostoc citreum</i>	99.07	96.03	0.02	YES	felino_salami
miao_suanrou.6	<i>Leuconostoc kimchii</i>	99.43	80.06	0.4	NO	miao_suanrou
miao_suanrou.46	<i>Leuconostoc pseudomesenteroides</i>	97.57	60.89	1.12	NO	miao_suanrou
miao_suanrou.35	<i>Furfurilactobacillus milii</i>	99.54	73.85	5.44	NO	miao_suanrou
miao_suanrou.47	<i>Furfurilactobacillus rossiae</i>	97.39	81.55	3.37	NO	miao_suanrou
miao_suanrou.7	<i>Furfurilactobacillus sp. (uSGB)</i>	N/A	61.97	0.04	NO	miao_suanrou
piemonte_salami.14	<i>Companilactobacillus versmoldensis</i>	99.86	89.87	0	YES	DFS_chorizo/DFS_salchichon/piemonte_salami
piemonte_salami.7	<i>Companilactobacillus mishanensis</i>	99.91	99.67	0.39	YES	piemonte_salami
miao_suanrou.73	<i>Streptococcus sp. (uSGB)</i>	N/A	94.15	0.18	NO	miao_suanrou
miao_suanrou.81	<i>Lactocaseibacillus paracasei</i>	98.67	98.8	0.43	NO	miao_suanrou
miao_suanrou.57	<i>Psychrobacter arenosus</i>	96.71	59.64	2.18	NO	miao_suanrou
mutton_sausage.1	<i>Vibrio hibernica</i>	98.76	99.34	0.18	YES	mutton_sausage
cured_tongue.24	<i>Psychrobacter sanguinis</i>	97.59	59.68	0.03	YES	cured_tongue
dry_aged_beef.31	<i>Psychrobacter pasteurii</i>	97.09	94.35	4.25	YES	dry_aged_beef
DFS_chorizo.11	<i>Acinetobacter harbinensis</i>	98.53	95	2.62	YES	DFS_chorizo/dry_aged_beef
dry_aged_beef.2	<i>Psychrobacter cibarius_A</i>	97.57	62.01	0.42	YES	DFS_chorizo/cecina/dry_aged_beef/miao_suanrou
cecina.54	<i>Psychrobacter immobilis_F</i>	97.15	73.96	0.87	YES	DFS_chorizo/cecina/dry_aged_beef
miao_suanrou.13	<i>Psychrobacter sp. (uSGB)</i>	N/A	61.04	7.6	NO	miao_suanrou
miao_suanrou.38	<i>Psychrobacter alimentarius</i>	97.83	50.55	1.22	NO	miao_suanrou
dry_aged_beef.15	<i>Bacillus velezensis</i>	98.8	96.91	0.57	YES	DFS_chorizo/dry_aged_beef
cecina.64	<i>Specibacter sp. (uSGB)</i>	N/A	96.27	0.99	YES	cecina
miao_suanrou.122	<i>Rothia endophytica</i>	98.47	99.9	1.88	NO	miao_suanrou
cured_tongue.13	<i>Kocuria salsicia</i>	96.01	89	4.41	YES	cured_tongue
miao_suanrou.28	<i>Kocuria rhizophila_A</i>	99.13	99.99	4.09	NO	miao_suanrou
cured_tongue.21	<i>Rothia sp. (uSGB)</i>	N/A	54.85	1.22	YES	cured_tongue
cured_tongue.3	<i>Dermaococcus nishinomiyaensis</i>	98.83	68.71	2.19	YES	cured_tongue
cured_tongue.30	<i>Brevibacterium sp. (uSGB)</i>	N/A	75.91	3.16	YES	cured_tongue
miao_suanrou.30	<i>Glutamicibacter bergerei</i>	98.48	82.66	1.43	NO	miao_suanrou
cecina.20	<i>Aquipseudomonas shahriarae</i>	98.8	66.68	3	YES	cecina
cecina.71	<i>Aquipseudomonas weihenstephanensis</i>	96.87	71.66	8.05	YES	cecina
miao_suanrou.56	<i>Serratia proteamaculans</i>	97.18	59.77	1.42	NO	miao_suanrou
miao_suanrou.60	<i>Serratia proteamaculans_D</i>	95.44	67.53	0.6	NO	miao_suanrou
miao_suanrou.97	<i>Serratia liquefaciens</i>	98.56	80.34	2.56	NO	miao_suanrou
miao_suanrou.39	<i>Erwinia sp. (uSGB)</i>	N/A	95.35	4.12	NO	miao_suanrou
miao_suanrou.45	<i>Hafnia alvei</i>	95.64	56.99	1.12	NO	miao_suanrou

MAG_representative	Assigned_taxon_GTDB	ANI_%	Completeness_%	Contamination_%	In_core_wildtype	Product(s)
miao_suanrou.33	<i>Pseudocitrobacter corydidari</i>	95.51	88.64	2.91	NO	miao_suanrou
miao_suanrou.90	<i>Kluyvera intermedia</i>	99.03	92.41	7.23	NO	miao_suanrou
miao_suanrou.15	<i>Klebsiella terrigena</i>	96.52	95.41	3.48	NO	miao_suanrou
miao_suanrou.130	<i>Dryocola sp051307055</i>	98.91	70.57	1.72	NO	miao_suanrou
miao_suanrou.70	<i>Buttiauxella gaviniae</i>	98.56	93.87	4.07	NO	miao_suanrou
miao_suanrou.3	<i>Pantoea floridensis</i>	98.87	85.68	3.35	NO	miao_suanrou
miao_suanrou.131	<i>Pantoea astica</i>	99.2	80.38	1.61	NO	miao_suanrou
miao_suanrou.78	<i>Pantoea brenneri</i>	99.56	88.19	4.12	NO	miao_suanrou
miao_suanrou.53	<i>Pantoea agglomerans</i>	97.66	85.71	1.42	NO	miao_suanrou
miao_suanrou.121	<i>Rahnella perminowiae</i>	99.09	91.08	4.85	NO	miao_suanrou
miao_suanrou.48	<i>Rahnella aceris</i>	99.34	78.13	1.27	NO	miao_suanrou
miao_suanrou.52	<i>Rahnella inusitata</i>	99.01	91.18	4.4	NO	miao_suanrou
cured_tongue.17	<i>Ewingella americana</i>	97.52	99.08	0.01	YES	cured_tongue
miao_suanrou.26	<i>Schleiferilactobacillus harbinensis</i>	98.53	90.23	1.34	NO	miao_suanrou

Appendix C.2 - Table 3B. Eukaryotic operational genomic groups

MAG_representative	Preliminary_ID	Completeness_%	Contamination_%	WT_status	In_core_wildtype	Product(s)
DFS_chorizo.6	Near the FastANI ~80% floor: assignment reliable only at order level (<i>Mucorales</i>); closest reference <i>Mucor velutinosus</i> NIH1002 but species unresolved - ANI 79.66% (reference GCF_035048745.1), aligned fraction 2070/3659 (~57%) Intermediate operational category: contamination 6.52% >5% (possible mixed bin; interpret species assignment with caution)	50.43	6.52	wild_type	YES	DFS_chorizo
DFS_chorizo.9	Near the FastANI ~80% floor: assignment reliable only at order level (<i>Mucorales</i>); closest reference <i>Mucor velutinosus</i> NIH1002 but species unresolved - ANI 79.64% (reference GCF_035048745.1), aligned fraction 2146/3667 (~59%) Intermediate operational category: contamination 7.39% >5% (possible mixed bin; interpret species assignment with caution)	53.91	7.39	wild_type	YES	DFS_chorizo
merged.0	No FastANI hit (no reference above the ~80% floor); retained as <i>Debaryomycetaceae</i> (yeasts) - preliminary assignment by EukCC	85.51	0.00	wild_type	YES	DFS_salchichon/ merged
dry_aged_beef.12	Closest to <i>Penicillium verrucosum</i> IBT 35672 - ANI 97.58% (reference GCF_028828655.1), aligned fraction 6771/7537	95.86	0.00	wild_type	YES	DFS_salchichon/ dry_aged_beef
cured_tongue.14	Closest to <i>Penicillium solitum</i> IBT 25940 - ANI 99.53% (reference GCF_028829755.1), aligned fraction 6257/6615	85.71	0.75	wild_type	YES	cecina/cured_tongue
DFS_chorizo.24	Closest to <i>Penicillium solitum</i> IBT 25940 - ANI 93.75% (reference GCF_028829755.1), aligned fraction 8813/10247; Species assignment preliminary (ANI <95%; fungal species threshold not established; genus/family assignment reliable)	100	0.38	wild_type	YES	DFS_chorizo
DFS_chorizo.3	Closest to <i>Debaryomyces hansenii</i> CBS767 - ANI 99.83% (reference GCF_000006445.2), aligned fraction 3836/3872	99.38	0.21	wild_type	YES	DFS_chorizo/DFS_salchichon/dry_aged_beef

MAG_representative	Preliminary_ID	Completeness_%	Contamination_%	WT_status	In_core_wildtype	Product(s)
cecina.7	Closest to <i>Debaryomyces hansenii</i> CBS767 - ANI 91.17% (reference GCF_000006445.2), aligned fraction 1994/2147; Species assignment preliminary (ANI <95%; fungal species threshold not established; genus/family assignment reliable)	62.11	2.28	wild_type	YES	cecina
cecina.31	Closest to <i>Debaryomyces hansenii</i> CBS767 - ANI 84.55% (reference GCF_000006445.2), aligned fraction 2789/3076; Species assignment preliminary (ANI <95%; fungal species threshold not established; genus/family assignment reliable)	89.23	0.41	wild_type	YES	cecina
DFS_salchichon.24	Closest to <i>Aspergillus glaucus</i> CBS 516.65 - ANI 92.96% (reference GCF_001890805.1), aligned fraction 7306/8256; Species assignment preliminary (ANI <95%; fungal species threshold not established; genus/family assignment reliable)	98.5	0.00	wild_type	YES	DFS_salchichon/cecina
cured_tongue.8	Closest to <i>Aspergillus glaucus</i> CBS 516.65 - ANI 97.44% (reference GCF_001890805.1), aligned fraction 5698/6345 Intermediate operational category: contamination 5.26% >5% (possible mixed bin; interpret species assignment with caution)	76.32	5.26	wild_type	YES	cured_tongue
miao_suanrou.85	Closest to <i>Saccharomyces cerevisiae</i> S288C - ANI 99.38% (reference GCF_000146045.2), aligned fraction 3668/3683 Inoculated Miao external group, excluded from the core	100	0.00	inoculated_Miao_external_group	NO	miao_suanrou
cecina.3	No FastANI hit (no reference above the ~80% floor); retained as <i>Debaryomycetaceae</i> (yeasts) - preliminary assignment by EukCC	76.81	0.21	wild_type	YES	cecina
cecina.22	OFF-TARGET: protist (<i>Kipferlia</i> (protist off-target)); no_hit FastANI (no reference below floor ~80%); EXCLUDED from the wild-type core	71.43	1.59	off_target_protista (<i>Kipferlia</i>)	NO	cecina
cecina.52	No FastANI hit (no reference above the ~80% floor); retained as <i>Dipodascaceae</i> (yeasts) - preliminary assignment by EukCC	87.16	0.00	wild_type	YES	cecina

Appendix C.3 - Table 3C. Curated wild-type reference set

Kingdom	Representative_genome	Product(s)	Completeness_%	Contamination_%	Note
prokaryote	cured_tongue.9	cured_tongue/miao_suanrou/mutton_sausage	78.4	0.14	Intermediate operational category: completeness 78.40% -> interpret with caution
prokaryote	cured_tongue.20	DFS_salchichon/cured_tongue	99.22	0.18	
prokaryote	felino_salami.8	cecina/dry_aged_beef/felino_salami/miao_suanrou/mutton_sausage	92.22	0.17	
prokaryote	dry_aged_beef.33	dry_aged_beef	88.96	0.15	Intermediate operational category: completeness 88.96% -> interpret with caution
prokaryote	felino_salami.15	felino_salami	96.75	1.29	DB representative for cluster 6_1 (override: dRep representative was Miao) Intermediate operational category: completeness 60.60% -> interpret with caution
prokaryote	felino_salami.4	felino_salami/miao_suanrou	60.6	1.21	
prokaryote	dry_aged_beef.4	DFS_chorizo/dry_aged_beef	94.67	1.22	
prokaryote	piemonte_salami.23	miao_suanrou/piemonte_salami	95.43	0.01	
prokaryote	DFS_salchichon.21	DFS_salchichon/felino_salami/miao_suanrou/piemonte_salami	99.99	0.19	

Kingdom	Representative_genome	Product(s)	Completeness _%	Contaminatio n_%	Note
prokaryote	DFS_chorizo.50	DFS_chorizo	78.04	9.83	Intermediate operational category: contamination 9.83% >5%; completeness 78.04% -> interpret with caution
prokaryote	DFS_chorizo.45	DFS_chorizo	98.71	3.3	
prokaryote	dry_aged_beef.30	DFS_chorizo/cecina/dry_aged_beef/felino_salami/miao_suanrou	98.61	0.05	
prokaryote	piemonte_salami.24	piemonte_salami	86.39	1.26	Intermediate operational category: completeness 86.39% -> interpret with caution
prokaryote	cured_tongue.28	DFS_salchichon/ciauscolo/cured_tongue/felino_salami/miao_suanrou/piemonte_salami	99.65	0.36	
prokaryote	ciauscolo.5	DFS_chorizo/DFS_salchichon/ciauscolo/felino_salami/piemonte_salami	87.06	0.21	Intermediate operational category: completeness 87.06% -> interpret with caution
prokaryote	ciauscolo.4	ciauscolo	100	0.58	
prokaryote	DFS_salchichon.14	DFS_salchichon	93.55	0.17	
prokaryote	DFS_salchichon.18	DFS_chorizo/DFS_salchichon	99.98	0.08	Intermediate operational category: fails GUNC (possible chimerism); completeness 83.03% -> interpret with caution
prokaryote	cured_tongue.5	cured_tongue	99.13	0.19	
prokaryote	felino_salami.20	felino_salami	83.03	3.41	
prokaryote	felino_salami.7	felino_salami/miao_suanrou	95.37	2.06	Intermediate operational category: contamination 5.83% >5% -> interpret with caution
prokaryote	piemonte_salami.18	felino_salami/piemonte_salami	99.93	0.06	
prokaryote	DFS_chorizo.48	DFS_chorizo/miao_suanrou	93.47	5.83	
prokaryote	felino_salami.12	felino_salami	94.78	7.47	Intermediate operational category: contamination 7.47% >5% -> interpret with caution
prokaryote	felino_salami.5	felino_salami	96.03	0.02	
prokaryote	piemonte_salami.14	DFS_chorizo/DFS_salchichon/piemonte_salami	89.87	0.00	Intermediate operational category: completeness 89.87% -> interpret with caution
prokaryote	piemonte_salami.7	piemonte_salami	99.67	0.39	
prokaryote	mutton_sausage.1	mutton_sausage	99.34	0.18	
prokaryote	cured_tongue.24	cured_tongue	59.68	0.03	Intermediate operational category: completeness 59.68% -> interpret with caution
prokaryote	dry_aged_beef.31	dry_aged_beef	94.35	4.25	
prokaryote	DFS_chorizo.11	DFS_chorizo/dry_aged_beef	95	2.62	Intermediate operational category: completeness 62.01% -> interpret with caution
prokaryote	dry_aged_beef.2	DFS_chorizo/cecina/dry_aged_beef/miao_suanrou	62.01	0.42	
prokaryote	cecina.54	DFS_chorizo/cecina/dry_aged_beef	73.96	0.87	Intermediate operational category: completeness 73.96% -> interpret with caution
prokaryote	dry_aged_beef.15	DFS_chorizo/dry_aged_beef	96.91	0.57	
prokaryote	cecina.64	cecina	96.27	0.99	Intermediate operational category: completeness 89.00% -> interpret with caution
prokaryote	cured_tongue.13	cured_tongue	89	4.41	
prokaryote	cured_tongue.21	cured_tongue	54.85	1.22	Intermediate operational category: completeness 54.85% -> interpret with caution

Kingdom	Representative_genome	Product(s)	Completeness _%	Contaminatio n_%	Note
prokaryote	cured_tongue.3	cured_tongue	68.71	2.19	Intermediate operational category: completeness 68.71% -> interpret with caution
prokaryote	cured_tongue.30	cured_tongue	75.91	3.16	Intermediate operational category: completeness 75.91% -> interpret with caution
prokaryote	cecina.20	cecina	66.68	3	Intermediate operational category: completeness 66.68% -> interpret with caution
prokaryote	cecina.71	cecina	71.66	8.05	Intermediate operational category: contamination 8.05% >5%; completeness 71.66% -> interpret with caution
prokaryote	cured_tongue.17	cured_tongue	99.08	0.01	<i>Mucorales</i> (moulds) Intermediate operational category: contamination 6.52% >5%; completeness 50.43% -> interpret with caution
eukaryote	DFS_chorizo.6	DFS_chorizo	50.43	6.52	
eukaryote	DFS_chorizo.9	DFS_chorizo	53.91	7.39	<i>Mucorales</i> (moulds) Intermediate operational category: contamination 7.39% >5%; completeness 53.91% -> interpret with caution
eukaryote	merged.0	DFS_salchichon/merged	85.51	0.00	<i>Debaryomycetaceae</i> (yeasts) Intermediate operational category: completeness 85.51% -> interpret with caution
eukaryote	dry_aged_beef.12	DFS_salchichon/dry_aged_beef	95.86	0.00	<i>Penicillium</i> (moulds)
eukaryote	cured_tongue.14	cecina/cured_tongue	85.71	0.75	<i>Penicillium</i> (moulds) Intermediate operational category: completeness 85.71% -> interpret with caution
eukaryote	DFS_chorizo.24	DFS_chorizo	100	0.38	<i>Penicillium</i> (moulds)
eukaryote	DFS_chorizo.3	DFS_chorizo/DFS_salchichon/dry_aged_beef	99.38	0.21	<i>Debaryomycetaceae</i> (yeasts)
eukaryote	cecina.7	cecina	62.11	2.28	<i>Debaryomycetaceae</i> (yeasts) Intermediate operational category: completeness 62.11% -> interpret with caution
eukaryote	cecina.31	cecina	89.23	0.41	<i>Debaryomycetaceae</i> (yeasts) Intermediate operational category: completeness 89.23% -> interpret with caution
eukaryote	DFS_salchichon.24	DFS_salchichon/cecina	98.5	0.00	<i>Aspergillus</i> (moulds)
eukaryote	cured_tongue.8	cured_tongue	76.32	5.26	<i>Aspergillus</i> (moulds) Intermediate operational category: contamination 5.26% >5%; completeness 76.32% -> interpret with caution
eukaryote	cecina.3	cecina	76.81	0.21	<i>Debaryomycetaceae</i> (yeasts) Intermediate operational category: completeness 76.81% -> interpret with caution
eukaryote	cecina.52	cecina	87.16	0.00	<i>Dipodascaceae</i> (yeasts) Intermediate operational category: completeness 87.16% -> interpret with caution

Appendix C.4 - Table 3D. Summary counts

Metric

Dereplicated prokaryotic representatives (95% ANI operational clusters)

Dereplicated eukaryotic representatives (operational genomic groups)

of which FastANI hit

of which no_hit (below ~80% floor)

Core wild-type genomes (prokaryote)

Core wild-type genomes (eukaryote)

Core wild-type TOTAL

Prokaryote novelty: kSGB (known SGB)

Prokaryote novelty: uSGB (candidate unclassified SGB; novelty unconfirmed)

Prokaryote functional groups

Lactic acid bacteria (LAB)

Coagulase-negative staphylococci (CNS)

Surface actinobacteria

Psychrotrophic Gram-neg spoilage

Enterobacteria (Gram-neg, hygiene/spoilage)

Other *Firmicutes*

Prokaryote phyla

Bacillota

Pseudomonadota

Actinomycetota

Appendix D - Table 4. Taxa in the comparison between PGI Ciauscolo shotgun profiles and previous literature

Each taxon is listed once. The comparison is restricted to the four samples produced within the PGI area. Five literature-reported taxa detected only in the two out-of-area samples (*Acinetobacter johnsonii*, *Leuconostoc carnosum*, *Mammaliococcus sciuri*, *Pseudomonas fragi* and *Weissella cibaria*) are not listed. Subspecies records (e.g., *Lactobacillus sakei* subsp. *sakei*) were kept as reported in the sources.

Shared by PGI shotgun and literature (n=15)	Detected only by PGI shotgun (n=31)	Reported only in previous literature (n=47)
<i>Carnobacterium divergens</i>	<i>Acidipropionibacterium acidipropionici</i>	<i>Acinetobacter</i> sp.
<i>Debaryomyces hansenii</i>	<i>Aspergillus chevalieri</i>	<i>Alternaria tenuissima</i>
<i>Enterococcus faecalis</i>	<i>Chroococcidiopsis</i> sp. CCME 29	<i>Aspergillus</i> sp.
<i>Escherichia coli</i>	<i>Citrobacter freundii</i>	<i>Aureobasidium</i> sp.
<i>Lactocaseibacillus paracasei</i>	<i>Edwardsiella piscicida</i>	<i>Burkholderia</i> sp.
<i>Lactiplantibacillus paraplantarum</i>	<i>Enterobacter bugandensis</i>	<i>Candida galli</i>
<i>Lactiplantibacillus plantarum</i>	<i>Enterobacter hormaechei</i>	<i>Candida psychrophila</i>
<i>Lactococcus lactis</i>	<i>Enterococcus avium</i>	<i>Carnobacterium</i> sp.
<i>Latilactobacillus curvatus</i>	<i>Enterococcus devriesei</i>	<i>Cladosporium cladosporioides</i>
<i>Latilactobacillus sakei</i>	<i>Enterococcus gilvus</i>	<i>Cladosporium</i> sp.
<i>Levilactobacillus brevis</i>	<i>Enterococcus raffinosus</i>	<i>Debaryomyces</i> sp.
<i>Pediococcus pentosaceus</i>	<i>Enterococcus saigonensis</i>	<i>Dellagialia algida</i>
<i>Staphylococcus saprophyticus</i>	<i>Erwinia persicina</i>	<i>Didymella negriana</i>
<i>Staphylococcus xylosus</i>	<i>Klebsiella grimontii</i>	<i>Enterobacteriaceae</i>
<i>Cutibacterium acnes</i> (reported in the literature as <i>Propionibacterium acnes</i>)	<i>Klebsiella quasipneumoniae</i>	<i>Gardnerella</i> sp.
	<i>Lactiplantibacillus pentosus</i>	<i>Glomus hyderabadensis</i>
	<i>Lactococcus cremoris</i>	<i>Gluconacetobacter</i> sp.
	<i>Lactococcus garvieae</i>	<i>Hanseniaspora uvarum</i>
	<i>Lactococcus petauri</i>	<i>Kurtzmaniella santamariae</i>
	<i>Lactococcus</i> sp. NH2-7C	<i>Kurtzmaniella zeylanoides</i>
	<i>Lactococcus</i> sp. bn62	<i>Lactocaseibacillus casei</i>
	<i>Pantoea agglomerans</i>	<i>Lactobacillus delbrueckii</i>
	<i>Pantoea dispersa</i>	<i>Lactobacillus johnsonii</i>

Shared by PGI shotgun and literature (n=15)	Detected only by PGI shotgun (n=31)	Reported only in previous literature (n=47)
	<i>Pseudomonas simiae</i>	<i>Lactobacillus sakei</i> subsp. <i>sakei</i>
	<i>Rahnella aquatilis</i>	<i>Lactobacillus</i> sp.
	<i>Salinibacter ruber</i>	<i>Lactococcus lactis</i> subsp. <i>lactis</i>
	<i>Serratia liquefaciens</i>	<i>Lactococcus</i> sp.
	<i>Serratia odorifera</i>	<i>Leuconostoc mesenteroides</i>
	<i>Serratia rhizosphaerae</i>	<i>Malassezia globosa</i>
	<i>Serratia</i> sp. 1D1416	<i>Malassezia</i> sp.
	<i>Vagococcus fluvialis</i>	<i>Malassezia yamatoensis</i>
		<i>Pediococcus acidilactici</i>
		<i>Penicillium hirsutum</i>
		<i>Penicillium roqueforti</i>
		<i>Penicillium</i> sp.
		<i>Pseudomonas</i> sp.
		<i>Psychrobacter</i> sp.
		<i>Rhizopus oryzae</i>
		<i>Rhodotorula mucilaginosa</i>
		<i>Saccharomyces barnettii</i>
		<i>Staphylococcus epidermidis</i>
		<i>Staphylococcus equorum</i>
		<i>Staphylococcus</i> sp.
		<i>Streptococcus infantis</i>
		<i>Tausonia pullulans</i>
		<i>Weissella hellenica</i>
		<i>Yarrowia deformans</i>

Note. 'Detected only by PGI shotgun' means not reported in the selected literature and does not indicate novelty. 'Reported only in previous literature' does not demonstrate biological absence from the four PGI samples. Taxonomic synonyms were counted once.

Appendix E. Supplementary data for the literature synthesis

Table E1. Reporting frequencies of the taxa discussed in Chapter 2

Taxon	Analysis units with a recorded presence	Frequency (%)
<i>Latilactobacillus sakei</i>	124	44.0
<i>Staphylococcus</i> spp.	98	34.8
<i>Pseudomonas</i> spp.	74	26.2
<i>Staphylococcus xylosus</i>	72	25.5
<i>Staphylococcus equorum</i>	69	24.5
<i>Lactiplantibacillus plantarum</i>	66	23.4
<i>Latilactobacillus curvatus</i>	66	23.4

Percentages use all 282 analytical units as the denominator and are rounded to one decimal place. Each unit may contain several taxa, so percentages do not sum to 100%. These values measure reporting frequency, not relative abundance. Genus-level and species-level records retain the resolution of the source matrix. Multiple units may derive from the same publication and are not necessarily independent biological samples. Counts were obtained from the presence markers in the Microorganisms sheet of Table_S2_microbial_matrix.xlsx.

Table E2. Multivariate results for the literature synthesis

A. PERMANOVA with each factor fitted separately

Factor	R2	Pseudo-F	Permutation p
Continent	0.0335	3.1682	0.001
Product category	0.0450	3.2131	0.001
Analytical technique	0.0423	6.0703	0.001

B. Partitioning of variation in the complete model (278 analytical units)

Fraction	Adjusted R2	Variation (%)
Unique to continent	0.0132	1.32
Unique to product category	0.0243	2.43
Unique to analytical technique	0.0306	3.06
Residual	0.9220	92.20

Panel B lists the three unique fractions and the residual. Shared fractions are not displayed, so these rows do not sum to 100%. Adjusted R2 fractions in panel B differ from the unadjusted, single-factor R2 values in panel A and should not be interpreted as the same measure. The complete model retained 278 units after excluding the four North American units under the minimum group-size criterion of five. The residual describes variation not explained by this model; it does not estimate the proportion of unknown microorganisms.

C. Tests of multivariate dispersion (PERMDISP)

Grouping factor	F	Permutation p
Continent	6.3549	0.001
Product category	7.5336	0.001
Analytical technique	4.4406	0.016

All panels refer to 278 analytical units. Analyses used binary Jaccard distances. Dispersion differed among groups for all three factors, which requires caution when interpreting the PERMANOVA comparisons. Study structure and methodological heterogeneity also limit ecological interpretation. Values are transcribed from Table_S3_PERMANOVA_ANOSIM_varpart.xlsx, sheet Table_S3: rows 5-7 (panel A), 35-37 and 42 (panel B), and 29-31 (panel C). The statistical analyses were not rerun for this thesis revision.

Documentation of the co-occurrence network

The network considered the 50 most frequently reported taxa and tested 1,225 pairs using two-sided Fisher exact tests. Associations were retained when the Benjamini-Hochberg adjusted p-value was below 0.01 and the absolute phi coefficient exceeded 0.15. The edge table contains 309 retained associations: 304 positive and five negative. These are associations in the assembled literature records and do not establish biological interactions. The node and edge data are recorded in Table_S5_network_nodes.csv and Table_S6_network_edges_significant.csv.

Electronic supplementary files

The Supplementary_Review.zip archive, available through the QR code and repository link below, contains the study inventory, source presence/absence matrix, reporting-frequency table, network nodes and edges, extraction documentation and a verified extract of the complete-model statistical results. The README and SHA-256 manifest describe the files and their provenance. Source documentation contains unresolved differences in search descriptions and taxonomic grouping; the files are provided as project records, not as evidence that extraction accuracy or search completeness has been validated.



<https://github.com/GiampieroFederici/fermented-meat-thesis-supplementary>